Definition 8.3

If the actual value is not zero, then

$$\text{Relative error} = \frac{|\text{Actual value} - \text{Approximate value}|}{\text{Actual value}}$$

$$\text{Percentage error} = \text{Relative error} \times 100 \quad .$$

Note: Absolute error has unit of measurement where as relative error and percentage error are units free.

Note that, in the case of the above examples,

In the first case

The relative error $= \frac{1}{5} = 0.2$; and the percentage error $= \frac{1}{5} \times 100 = 20\%$.

In the second case

The relative error $= \frac{5}{100}$; and the percentage error $= \frac{5}{100} \times 100 = 5\%$.

So the second approximation is a better approximation than the first one. Note that, in order to calculate the relative error or the percentage error one should know the actual value of what we are approximating.

Let us consider some examples.

Example 8.3

Let us assume that the shape of a soap bubble is a sphere. Use linear approximation to approximate the increase in the surface area of a soap bubble as its radius increases from 5 cm to 5.2 cm. Also, calculate the percentage error.

Solution

Recall that surface area of a sphere with radius r is given by $S(r) = 4\pi r^2$. Note that even though we can calculate the exact change using this formula, we shall try to approximate the change using the linear approximation. So, using (4), we have

Change in the surface area

$$S(5.2) - S(5) \approx S'(5)(0.2)$$

$$= 8\pi(5)(0.2) = 8\pi \text{ cm}^2$$

Exact calculation of the change in the surface gives

$$S(5.2) - S(5) = 108.16\pi - 100\pi = 8.16\pi \text{ cm}^2 \quad .$$

$$\text{Percentage error} = \text{relative error} \times 100 = \frac{|8.16\pi - 8\pi|}{8.16\pi} \times 100 = 1.9607\%$$

Example 8.4

A right circular cylinder has radius $r = 10$ cm. and height $h = 20$ cm. Suppose that the radius of the cylinder is increased from 10 cm to 10.1 cm and the height does not change. Estimate the change in the volume of the cylinder. Also, calculate the relative error and percentage error.

Solution

Recall that volume of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height. So we have $V(r) = \pi r^2 h = 20\pi r^2$.

$$V(10.1) - V(10) \approx \left.\frac{dV}{dt}\right|_{r=10} (10.1 - 10) = 20\pi 2(10(0.1)) .$$

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